Project Title: Interactive Visual Analytics for Australian Hospitalised Road Traffic Injury Trends

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1.0 Introduction

1.1 Background and Motivation

The primary target audience for our interactive visualization includes the public health researches, transport safety policymakers, urban infrastructure planners and road advocacy groups. The platform also can be useful to everyday road users such as motorists, motorcyclists, cyclists, and pedestrians who might want to better understand road traffic risks in their community.

Road traffic injuries is among one of the major public health concern that Australia is currently facing which is placing a considerable pressure on emergency medical services which can result in a significant amount of hospitalisations every year. Even though government datasets are publicly available to everyone, traditional reports and static tables can make it much more difficult to identify patterns across road user types, age groups, and historical trends. Hence, in order to address this issue, the project aims to present a complex health and transport data through an accessible, interactive visualization that can allow users to explore the information much more easily.

The platform will let users to examine road safety trends and identify potentially high-risk groups. Users can explore the annual hospitalization numbers over an eleven-year period from 2011 to 2021 to identify any changes over time and investigate possible effects of major events such as the COVID-19 pandemic. They can also filter through the data by road user categories, including car drivers, pedal cyclists, and motorcyclists, and compare age-group distributions to better understand which population groups may face greater risks across different modes of transport.

1.2 Visualization Purpose

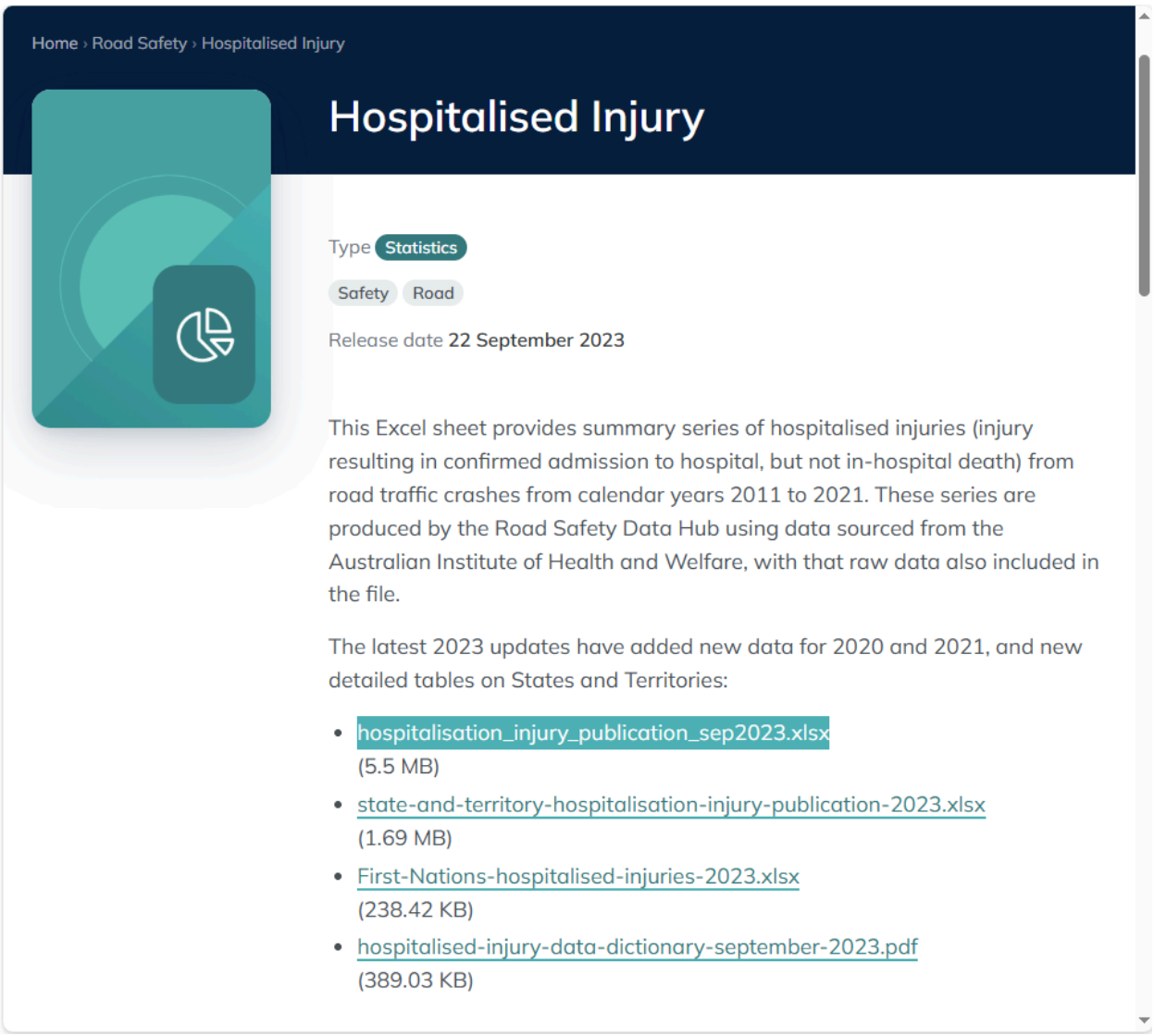
Our visualization aims to answer several important questions about road safety in Australia. It helps users identify which road user categories account for the highest number of hospitalised injuries and examine how injury cases changed between 2011 and 2021. The users can also compare injury patterns across different age groups and transport modes to identify potentially vulnerable populations. Moreover, our visualization will allow users to examine changes in hospitalisation numbers during 2020 when the quarantine for COVID-19 officially started which affected travel patterns and compare them with previous years.

By presenting the complex data through clear and interactive visualizations, the project can support users with a better decision-making based on evidence. Transport safety authorities can use the findings to develop targeted public education campaigns and review safety measures for vulnerable road users such as motorists and cyclists. Urban planners can also use the data to support decisions about investing in cycling paths, pedestrian crossings and other road safety infrastructure. Overall, the platform can make road safety data easier to explore and understand which can help stakeholders evaluate existing policies, identify areas for improvement and allocate resources more effectively to create safer roads across Australia.

2.0 Data

2.1 Data Source and Governance.

The primary data utilized in this project is sourced from the Bureau of Infrastructure and Transport Research Economics (BITRE) Hospitalised Injury Data (https://www.bitre.gov.au/resource/road-safety/hospitalised-injury) , published under the Australian Government Department of Infrastructure, Transport, Regional Development, Communications and the Arts.



Our Primary Publication File is from:
Hospitalisation_injury_publication_sep2023.xlsx.

These national hospitalisation series are compiled by the Road Safety Data Hub using underlying administrative health data curated by the Australian Institute of Health and Welfare (AIHW), capturing transport-related hospital admissions across calendar years 20011 through 2021.

Data Summary Table based on **“Hospitalisation_injury_publication_sep2023.xlsx.”**

Metadata Attribute	**Raw Source Dataset (raw sheet)**
Data Publisher / Provider	BITRE & Australian Institute of Health and Welfare (AIHW)
Temporal Coverage	Calendar Years 2011 – 2021 (Monthly granularity)
Geographic Scope	Australia-wide (ABS Remoteness Areas: Major Cities, Regional, Remote)
Target Population	Individuals admitted to hospital following transport-related incidents
Total Record Count (Rows)	117,919 rows
Total Column Count	13 attributes

Key Metrics / Measures	Hospitalisations, Count of cases, Bed days, Died_cases, Died_bed_days
Key Categorical Attributes	Calendar year, Month, ABS remoteness area, Cause of injury, Age group, Sex, Road user, Counterparty
Update Frequency	Irregular / Periodic annual releases (Latest update released September 2023)

(Refer back to Design Book Temp - Need add something more here,)

2.2 Data Processing and Analysis

3.0 Visualization Design

4.0 Iteration and Validation

5.0 Conclusion and Future Improvements